

CNSC Codex — Setup Guide

System Deployment Documentation Camarines Norte State College — Student Development Unit — Local Installation Reference

Table of Contents

- [CNSC Codex — Setup Guide](#)
 - [Table of Contents](#)
 - [1. Required Software and Tools to Install](#)
 - [2. Operating System Requirements](#)
 - [3. Database Backup File Location](#)
 - [4. Database Setup Instructions](#)
 - [Option A — Local MongoDB](#)
 - [Option B — MongoDB Atlas \(Cloud\)](#)
 - [5. Database Connection Credentials \(Local Setup\)](#)
 - [6. Port Configuration and Firewall Notes](#)
 - [7. Environment Configuration Files](#)
 - [File 1 — `server/.env`](#)
 - [File 2 — `client/.env`](#)
 - [8. Install and Run Commands](#)
 - [9. Local Access URL](#)
 - [10. Pre-Created Test Account Credentials](#)
 - [11. External Service Dependencies](#)
-

1. Required Software and Tools to Install

The following must be installed on the machine before running the project. No XAMPP, Docker, or PHP is needed — this is a fully Node.js and MongoDB stack.

- **Node.js v18 or higher** — the entire backend and frontend tooling depends on it. Download from nodejs.org
 - **npm v9 or higher** — included with Node.js; used for package management and running workspace scripts
 - **MongoDB** — either a local MongoDB Community Server installation, or a free cloud account on MongoDB Atlas at mongodb.com/atlas
 - **Git** — required to clone the repository from GitHub
 - **A Gmail account with App Password enabled** — required only for the email notification feature (Nodemailer). Standard Gmail passwords will not work if 2FA is active.
-

2. Operating System Requirements

There are no specific OS restrictions. The project is built entirely on Node.js, React (Vite), and MongoDB — all of which are cross-platform.

Compatible Systems Compatible with Windows, macOS, and Linux. There are no DLL dependencies or platform-specific binaries.

3. Database Backup File Location

This cannot be determined from the repository alone. The project uses **MongoDB** (not MySQL), so the backup format is a `.bson` dump folder (via `mongodump`) or a JSON export — not a `.sql` file. The uploads folder is gitignored and no backup file was committed to the repository.

⚠ Action Required Confirm with your group where the database dump was saved on the flash drive — for example: `/database/cnsc-codex-dump/` or `/database/cnsc_codex_backup.gz`. Also confirm the exact filename before the defense.

4. Database Setup Instructions

The system uses MongoDB. There are two possible scenarios depending on whether the database is local or cloud-hosted.

Option A — Local MongoDB

1. Install MongoDB Community Server from mongodb.com/try/download/community and start the service (it usually runs automatically after installation).
2. If you have a `mongodump` backup folder, run:

```
mongorestore --db cnsc-codex /path/to/dump/cnsc-codex
```

3. If you have a JSON export, use MongoDB Compass (free GUI) to import it into a database named `cnsc-codex`.
4. Set `MONGO_URI=mongodb://localhost:27017/cnsc-codex` in `server/.env`.

Option B — MongoDB Atlas (Cloud)

1. Create a free account at mongodb.com/atlas and create a cluster.
2. Create a database user with a username and password.
3. Get your connection string — it looks like:

```
mongodb+srv://username:password@cluster.mongodb.net/cnsc-codex
```

4. Paste that string as the value of `MONGO_URI` in `server/.env`.
5. No local import is needed if the live database is still active on Atlas — the system connects directly.

5. Database Connection Credentials (Local Setup)

Based on the README environment variable template, the default local configuration should be:

Setting	Value
MONGO_URI	mongodb://localhost:27017/cnsc-codex
Database Name	cnsc-codex
Database User	None (local MongoDB default)
Database Password	None (local MongoDB default)
Database Host	localhost
MongoDB Port	27017

⚠ Note If your group changed these credentials, check the actual `server/.env` file saved on the flash drive — it is gitignored and not visible in the GitHub repository.

6. Port Configuration and Firewall Notes

The system uses two ports that must be free and unblocked:

Port	Used By
5000	Express backend server
5173	Vite React frontend (development mode)

Instruction for Ma'am Please ensure ports 5000 and 5173 are free and not blocked by the Windows Firewall. If another application is already using port 5000, the backend will fail to start. No special firewall rules are needed beyond this, since everything runs on localhost.

7. Environment Configuration Files

There are two `.env` files that must be created manually. They are gitignored, so they will not be present after cloning — copy them from the flash drive or create them fresh.

File 1 — `server/.env`

```
# MongoDB connection
MONGO_URI=mongodb://localhost:27017/cnsc-codex

# Server
PORT=5000
SESSION_SECRET=any_random_secret_string_here

# Email (Gmail App Password)
EMAIL_USER=your_gmail@gmail.com
EMAIL_PASS=your_gmail_app_password

# Feature flags
```

```
ENABLE_REQUIREMENT_GATING=false  
REQUIREMENT_MAX_FILE_MB=10
```

File 2 — `client/.env`

```
VITE_API_URL=http://localhost:5000  
VITE_UPLOADS_URL=http://localhost:5000/uploads
```

⚠ **Switching from Live to Local Mode** If the system was previously pointed at a live cloud server, the `VITE_API_URL` in `client/.env` may be set to a live domain (e.g., `https://cnsc-codex.onrender.com`). To switch to local mode, change it to `http://localhost:5000`.

⚠ **Important** Make sure to include the actual `.env` files on the flash drive, since they contain real credentials that are not committed to GitHub.

8. Install and Run Commands

Open a terminal (Command Prompt or PowerShell on Windows; Terminal on Mac or Linux) and run the following commands in order:

```
# Step 1 – Navigate to the project folder  
cd CnscCodexMain  
  
# Step 2 – Install all dependencies  
# This installs both client and server packages at once via npm workspaces  
npm install  
  
# Step 3 – Start the application  
# This boots both the React frontend and Express backend simultaneously  
npm run dev
```

Note The `npm run dev` command uses `concurrently` to start both the backend and frontend at the same time. You do not need to open two separate terminals.

9. Local Access URL

Once the system is running, open a web browser and go to the following address:

Purpose	URL
Frontend (the actual website)	<code>http://localhost:5173</code>
Backend API (for testing only)	<code>http://localhost:5000</code>

Ma'am only needs to open `http://localhost:5173` in the browser to use and explore the system.

10. Pre-Created Test Account Credentials

No seed files or hardcoded test accounts were found in the repository. User accounts are stored in MongoDB and not committed to GitHub.

 **Action Required** Check your MongoDB database for existing accounts and note them down before the defense. Make sure at least one account per role is present in the database dump provided.

Role	Email	Password
SDU Administrator	<i>fill in</i>	<i>fill in</i>
SDU Coordinator	<i>fill in</i>	<i>fill in</i>
Dean	<i>fill in</i>	<i>fill in</i>
Adviser	<i>fill in</i>	<i>fill in</i>
Student Leader	<i>fill in</i>	<i>fill in</i>

11. External Service Dependencies

The system relies on several external services. The following table explains what may fail and how to handle it during the defense:

Feature	Dependency	What Happens Without It
Email notifications	Gmail via Nodemailer (requires internet + valid Gmail App Password in <code>.env</code>)	Accreditation email alerts will fail silently. The rest of the system continues to work normally.
Real-time notifications	Socket.IO (runs locally on port 5000 — no internet needed)	Works offline as long as the server is running.
File uploads	Stored locally in <code>server/uploads/</code>	Works offline as long as the folder exists locally.

Suggested Explanation for Ma'am The email notification feature and the AI document feedback feature require an active internet connection and valid API credentials. If those credentials are not configured, those two features will not work — but all core features including accreditation management, proposals, financial reports, and user management will function normally on localhost without an internet connection.